Course Code: Ethical Hacking Laboratory

Software Required: Kali Linux (Latest Version)

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Experiment 1

Basic File and User Management Commands

Aim

To execute and understand the usage of **rm, users, and tree** commands in Kali Linux.

Procedure

1. Open Kali Linux Terminal.
2. Create a sample file.
3. Execute the rm command.
4. Execute the users command.
5. Execute the tree command.

Commands:

Touch test.txt

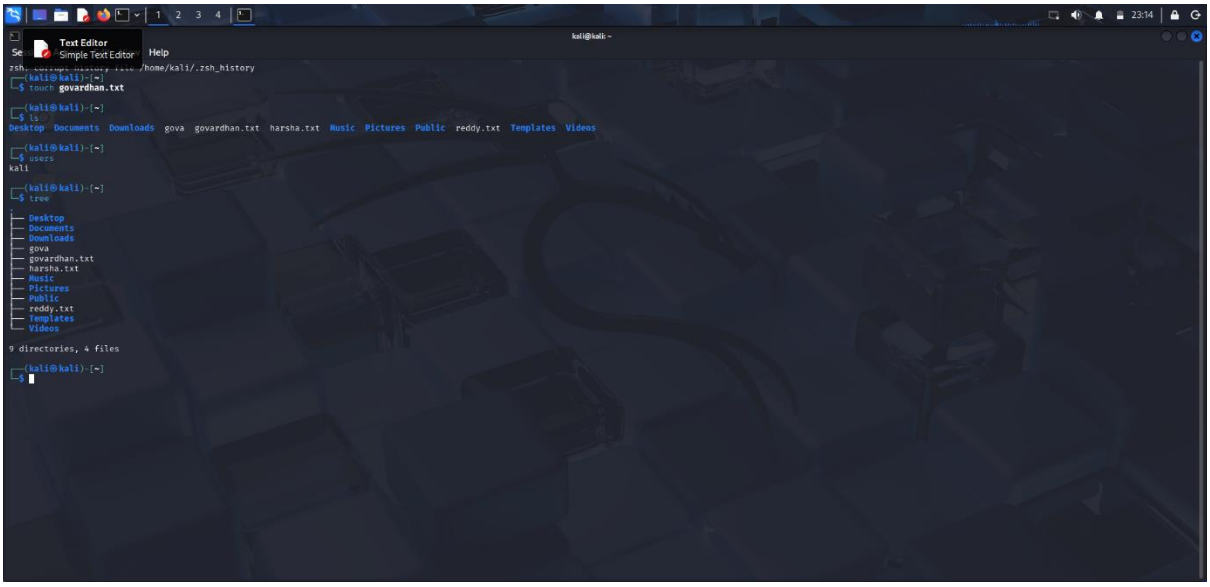
Ls

Rm test.txt

User

Tree

Output



Result

Successfully executed rm, users, and tree commands in Kali Linux.

Experiment 2

File Viewing Commands

Aim

To execute and understand the usage of **less, more and vi** commands.

Procedure

1. Create a text file.
2. Open the file using less.
3. Open the file using more.
4. Edit the file using vi.

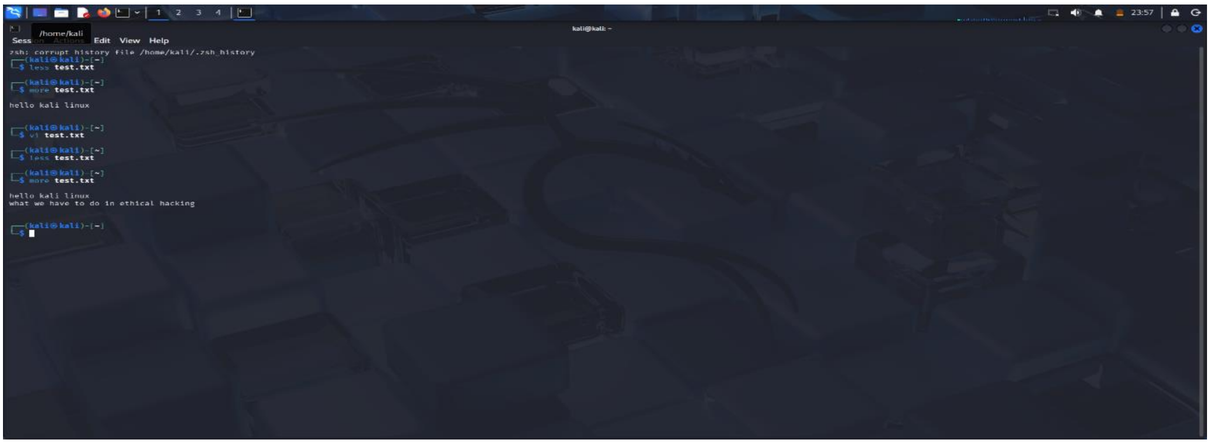
Commands:

less test.txt

more test.txt

vi test.txt

Output



Result

Successfully executed file viewing and editing commands.

Experiment 3

Tree, Head and Tail Commands

Aim

To execute and understand the usage of tree, head, and tail commands.

Procedure

1. Create a text file containing multiple lines.
2. Execute the head command.
3. Execute the tail command.
4. Display directory structure using tree.

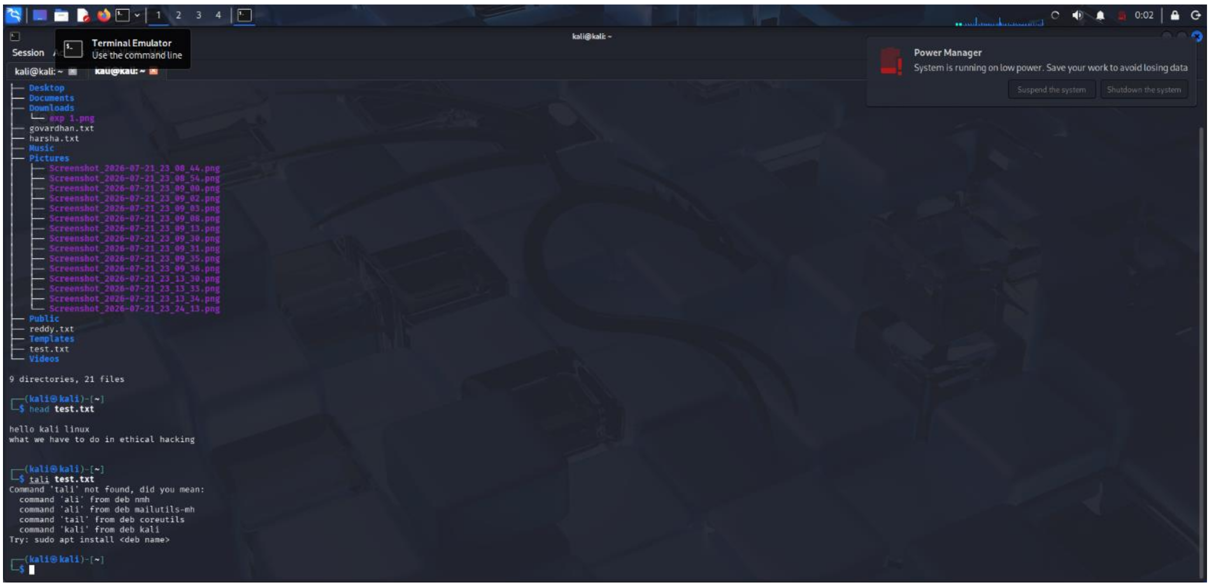
Commands:

Tree

Head test.txt

Tali test.txt

Output



Result

Successfully executed tree, head, and tail commands.

Experiment 4

Directory and File Display Commands

Aim

To execute ls, cat, and mkdir commands.

Procedure

1. Create a directory using mkdir.
2. List files using ls.
3. Create a text file.
4. Display file contents using cat.

Commands:

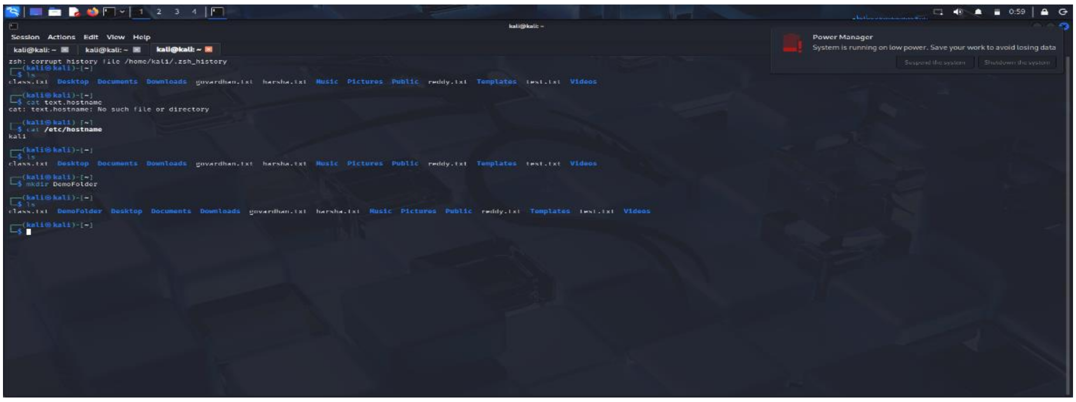
Ls

Cat /etc/hostname

Mkdir DemoFolder

ls

Output:



Result

Successfully executed ls, cat, and mkdir commands.

Experiment 5

System Information Commands

Aim

To execute **uptime, date, and mv** commands.

Procedure

1. Check system uptime.
2. Display current date.
3. Rename a file using mv.

Commands:

Uptime

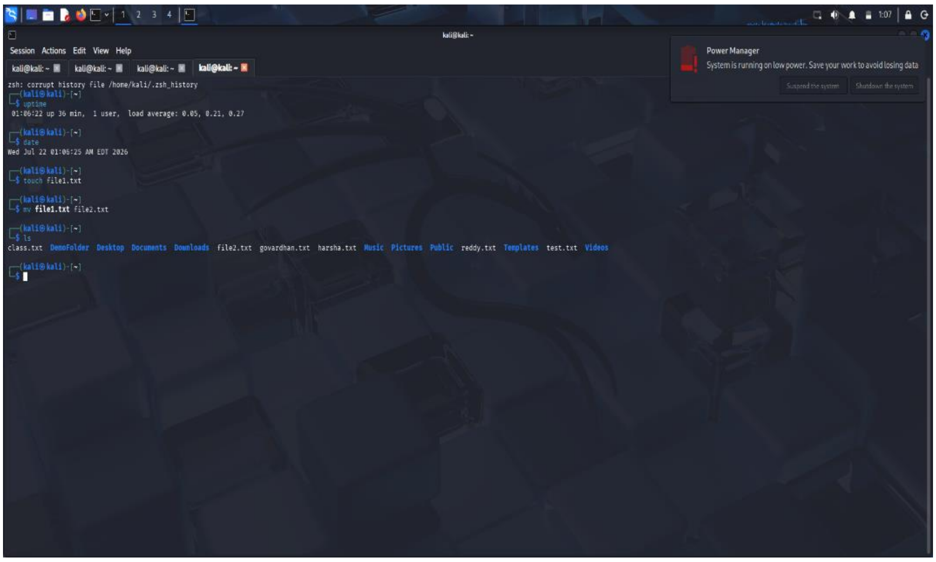
Date

Touch file1.txt

Mv file1.txt file2.txt

Ls

Output:



Result

Successfully executed uptime, date, and mv commands.

Experiment 6

File Creation and Download Commands

Aim

To execute touch, pwd, and wget commands.

Theory

touch Command

Creates an empty file.

touch sample.txt

pwd Command

Displays current working directory.

pwd

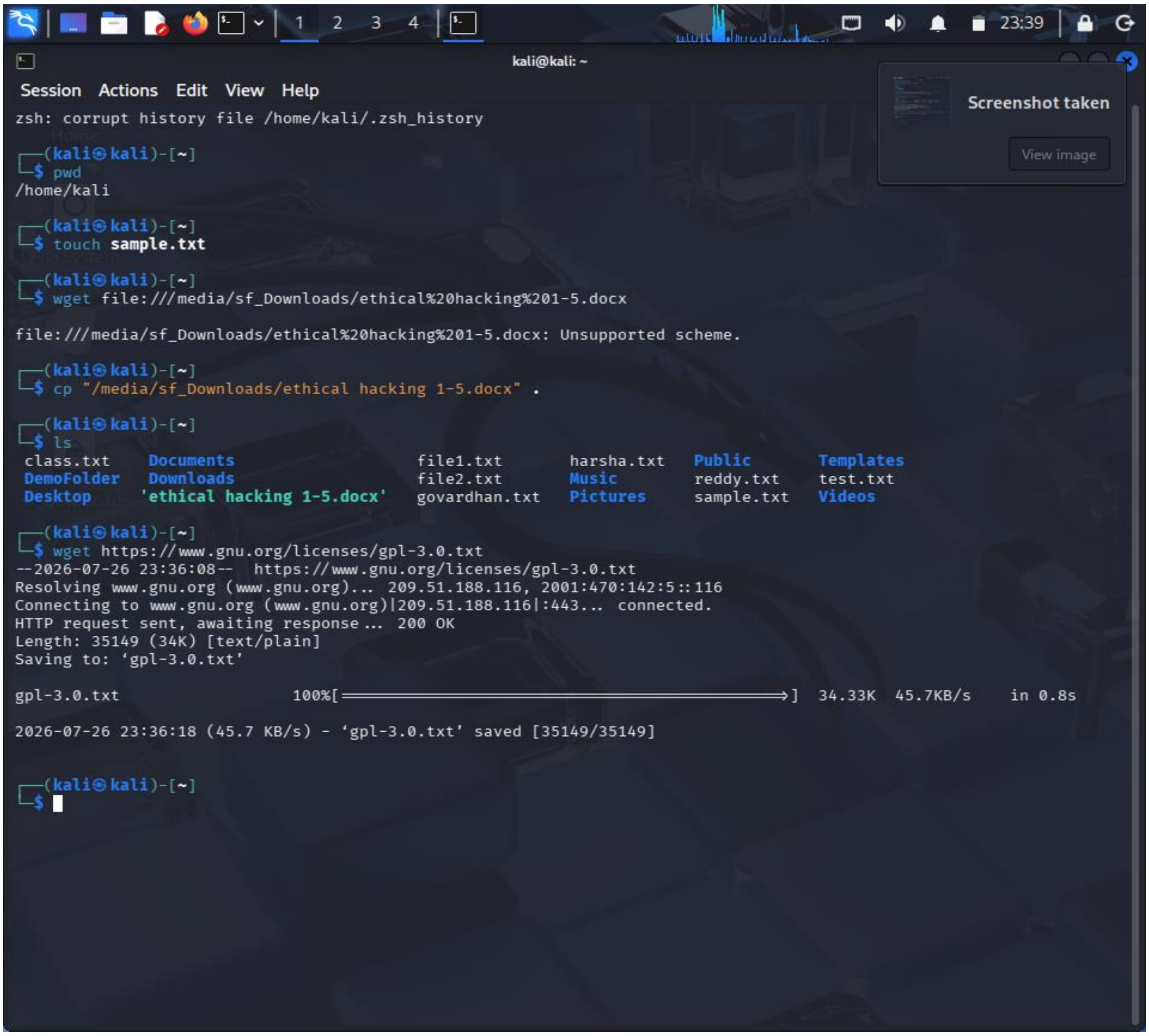
wget Command

Downloads files from the internet.

wget https://example.com/file.zip

Procedure

1. Create a file using touch.
2. Display current directory using pwd.
3. Download a sample file using wget.



Result

Successfully executed touch, pwd, and wget commands.

Experiment 7

Navigation and Copy Commands

Aim

To execute cd, cp, and ls commands.

Theory

cd Command

Changes the current directory.

cd Documents

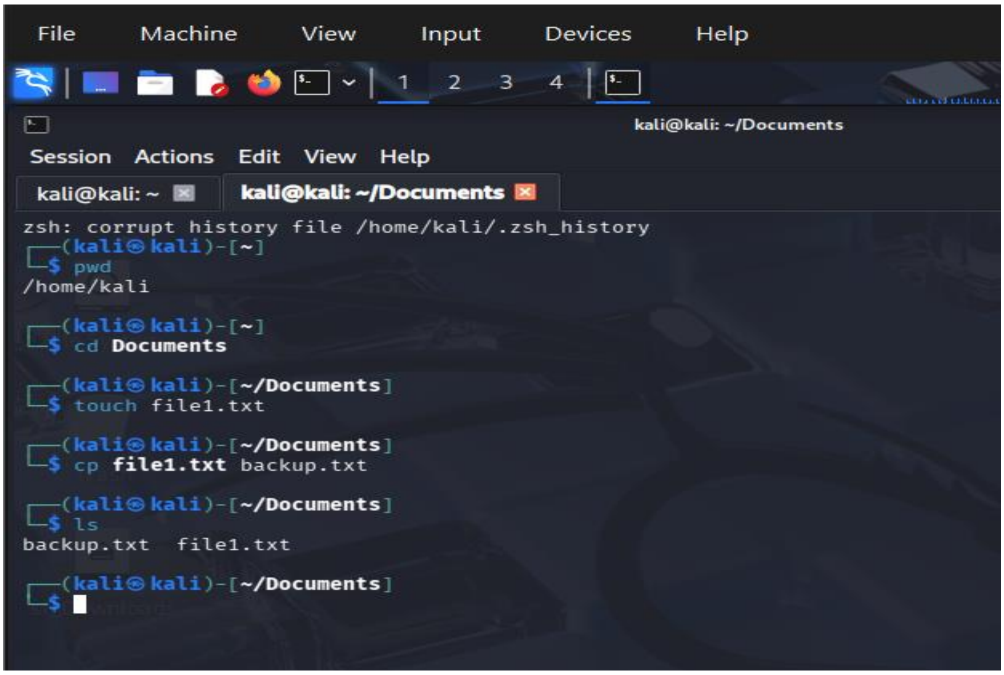
cp Command

Copies files and directories.

cp file1.txt backup.txt

Procedure

1. Navigate between directories using cd.
2. Copy files using cp.
3. Verify copied files using ls.



Result

Successfully executed cd, cp, and ls commands.

Experiment 8

Nmap Service Version and OS Detection

Aim

To perform service version detection and operating system identification using Nmap.

Theory

Nmap (Network Mapper) is an open-source network scanning tool used for network discovery and security auditing.

Commands

i) Detect Service Versions

nmap -sV 192.168.1.10

Description:
Determines the versions of services running on target ports.

ii) Aggressive Scan

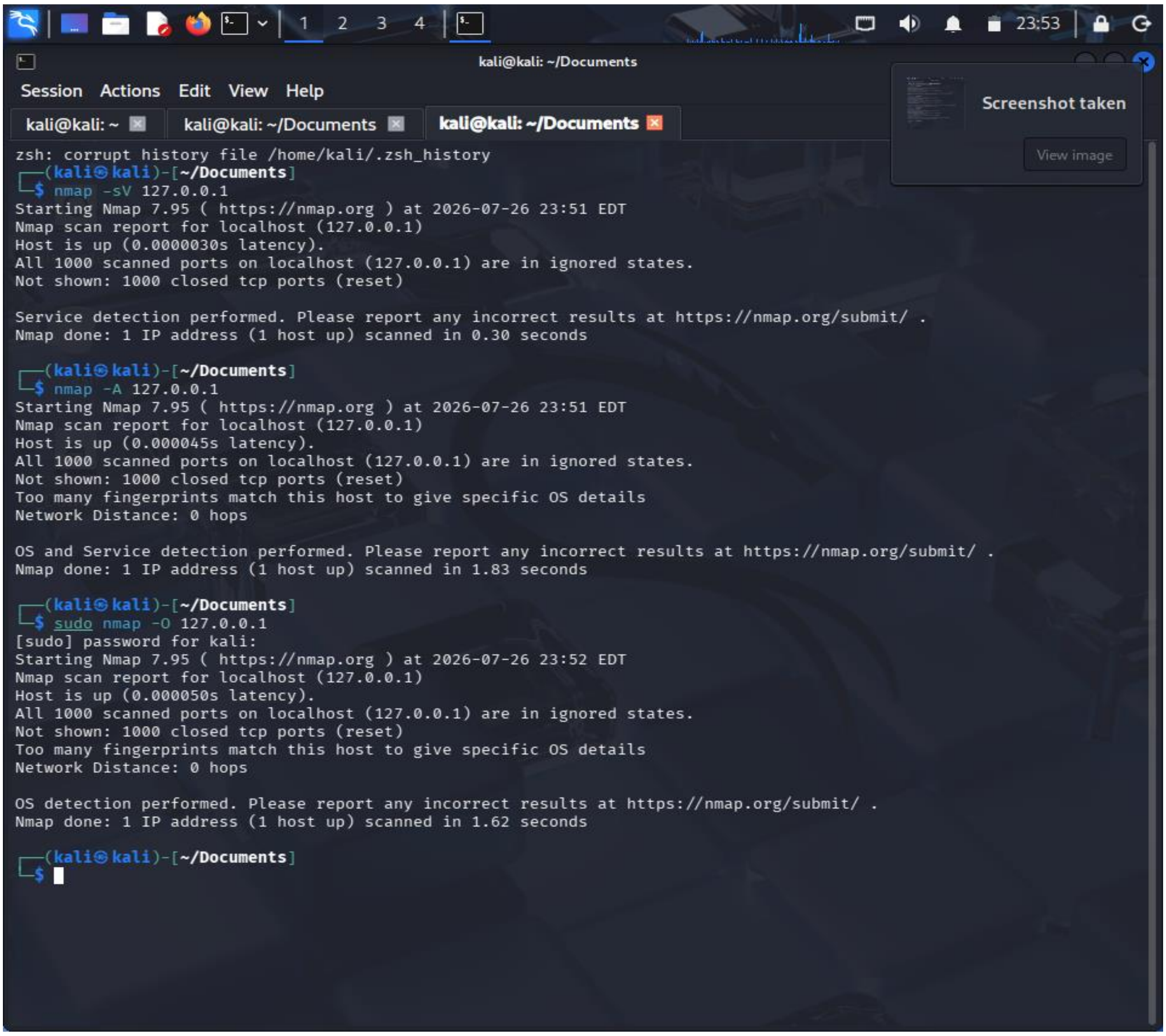
nmap -A 192.168.1.10

Description:
Performs OS detection, version detection, script scanning, and traceroute.

iii) Operating System Detection

sudo nmap -O 192.168.1.10

Description:
Identifies the operating system running on the target host.



Result

Successfully performed service version detection and OS identification using Nmap.

Experiment 9

Nmap Timing and Performance Commands

Aim

To perform IDS evasion and timing scans using Nmap.

Theory

Timing options control the speed and stealthiness of Nmap scans.

Commands

i) Polite IDS Evasion

nmap -T2 192.168.1.10

Description:
Slows down scanning to reduce the chance of detection.

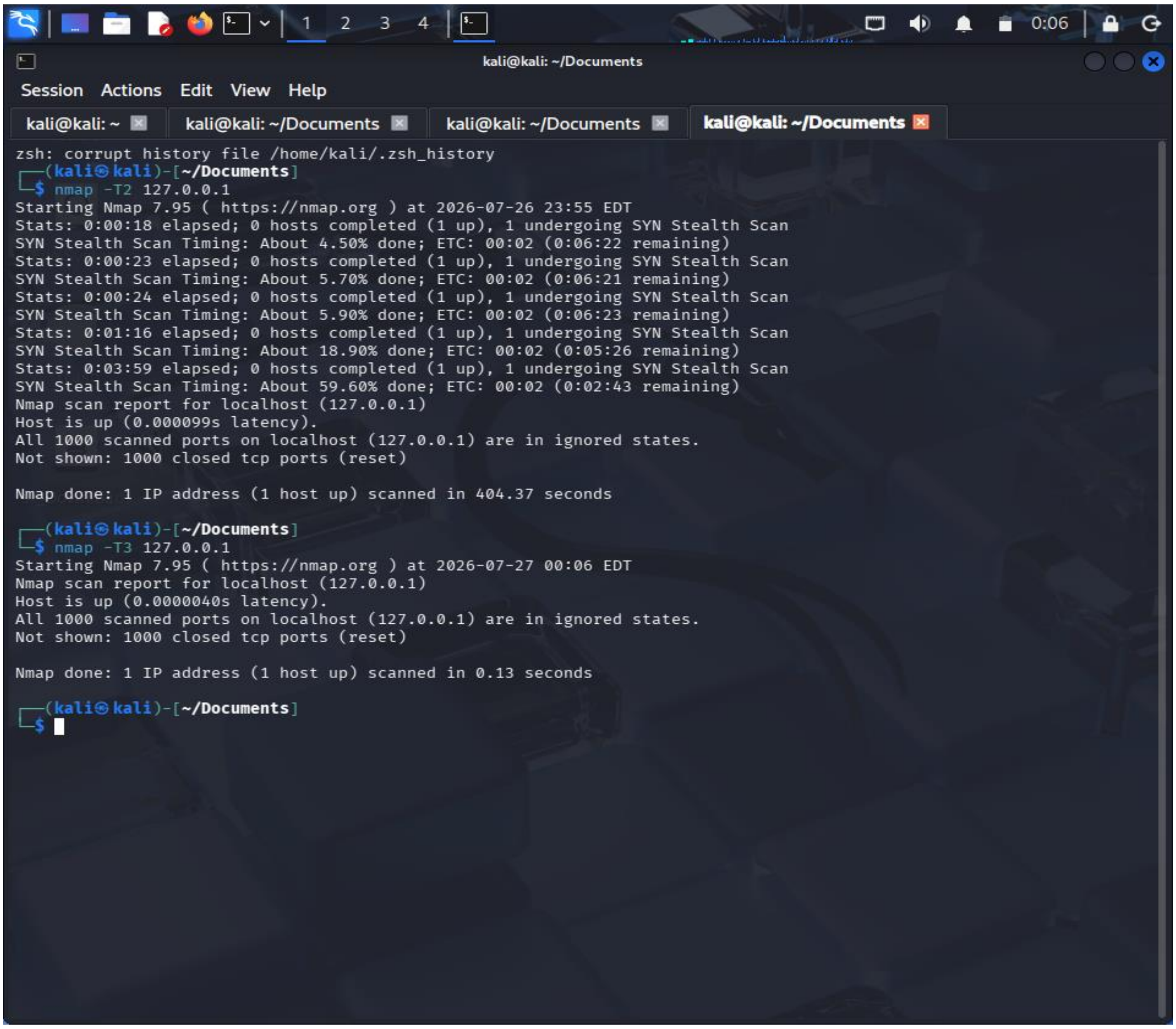
ii) Normal Scan

nmap -T3 192.168.1.10

Description:
Default balanced scanning speed.

Procedure

1. Identify target IP address.
2. Execute Nmap timing commands.
3. Observe scan duration and results.
4. Compare scan speeds.



Result

Successfully executed Nmap timing and performance scans in Kali Linux.

Experiment 10: Free, Sort and History Commands

Aim:
To execute and understand the usage of Free, Sort, and History commands in Kali Linux.

Commands:

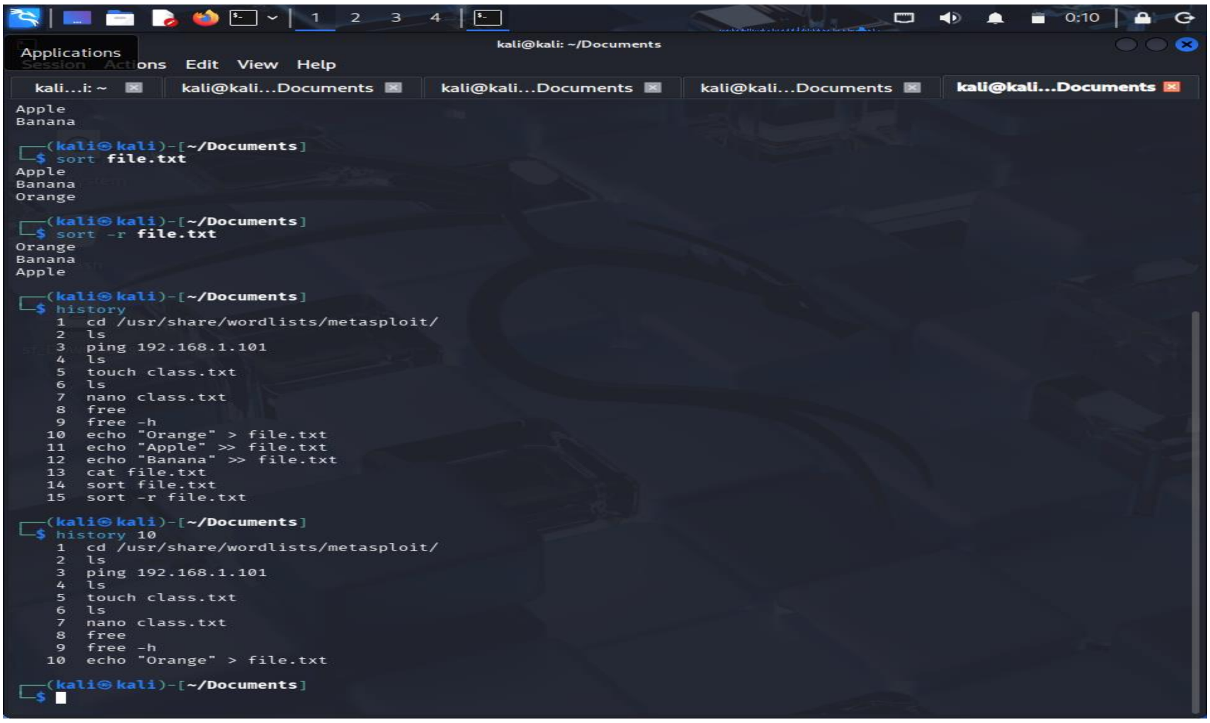
`free`
`free -h`

`sort file.txt`
`sort -r file.txt`

`history`
`history 10`

Procedure:

1. Open Terminal in Kali Linux.
2. Execute the `free` command to view memory usage.
3. Create a text file and use the `sort` command.
4. Execute the `history` command to display previously executed commands.



Result:
Successfully executed Free, Sort, and History commands in Kali Linux.